Pranav Rebala

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SUMMARY

AI engineer who builds and ships production LLM and agentic systems end to end, from prototype through deployment and monitoring, working forward-deployed and embedded with internal teams while delivering to external customers, pairing hands-on RAG, multi-agent, guardrail, and evaluation engineering with an MS in Business Analytics that keeps the work tied to real outcomes, and strong across Python, PyTorch, LangGraph, FastAPI, and AWS

TECHNICAL SKILLS

LLM & Agentic AI: LangChain, LangGraph, RAG, FAISS, Hugging Face, prompt engineering, evaluation and guardrails, multi-agent orchestration, MCP tool integration

Machine Learning: PyTorch, Scikit-learn, XGBoost, LightGBM, Prophet, NLP, recommendation systems, time-series forecasting

Engineering: Python, SQL, FastAPI, Flask, REST APIs, React, Node.js, PostgreSQL, MongoDB, SQLite

Cloud & Delivery: AWS (EC2, S3, Lambda, RDS), Azure, GCP Vertex AI, Docker, CI/CD, GitHub Actions, model monitoring

EXPERIENCE

Software Engineer, AI | General Aeronautics — Mar 2026 to Present

- Cut support resolution time from 48 hours toward a 3-hour target with an internal AI diagnostic tool that routes issues to the right one of 200+ verified repair procedures, gating the LLM behind deterministic routing so clear cases resolve without a model call
- Engineered schema-validated LLM output with hallucination guards that reject any category outside the retrieved set, and a guided-repair chat grounded in pre-extracted steps whose outcome feedback loop surfaces knowledge-base gaps back to engineering
- Built a Transformer and XGBoost hybrid pipeline classifying 58,479 MAVLink telemetry events across 3,718 flights into 40+ fault categories at 88.1% critical-event accuracy
- Built a multi-agent bug-fixing pipeline (planner, generator, evaluator) that autonomously diagnoses and patches defects, benchmarked against the real code shipped in historical commits

ML Engineer | Forfend Cybernatics | Remote — Jun 2025 to Mar 2026

- Deployed a collaborative-filtering recommendation engine into 15 external auto-dealership locations serving 20,000+ monthly users, with A/B-tested recommendation flows and ranking features that lifted engagement and conversion
- Shipped a RAG customer-service chatbot handling appointment booking, inventory lookups, and FAQ resolution, using retrieval evaluation and monitored fallback handling to hold answer quality at scale
- Built a LangGraph agentic workflow pairing Prophet and XGBoost demand forecasting with automated resource-allocation, deployed to AWS on Docker with real-time production monitoring

Full Stack Development Intern | Wipro Ltd | Bengaluru, India — Jul 2023 to Oct 2023

- Built a real-time financial reporting dashboard on Azure SQL and role-based access controls for 1,200+ users, cutting reporting lag from 2 days to 2 hours

Data Science Intern | Flutura Business Solutions | Bengaluru, India — Jun 2022 to Aug 2022

- Built predictive-maintenance models on IoT sensor data from 500+ industrial devices, reaching 85% fault-detection accuracy

PROJECTS

- **The Dead Collector's Estate (multi-agent mystery game):** architected a stateful five-character system in LangGraph with character-specific RAG stores, tool-calling flows, interrogation scoring, and persistent session memory (Python, LangGraph, OpenAI, RAG, FastAPI)
- **FootballBase (LLM sports analytics):** built a natural-language Q&A platform over 8,900+ football records with LangChain and FAISS semantic search, retrieval evaluation, and dynamic chart generation via MCP tool integration (Python, LangChain, FAISS, OpenAI, FastAPI)

EDUCATION

MS, Business Analytics | Babson College, Wellesley, MA — Aug 2024 to May 2025

BS, Computer Science (Machine Intelligence and Data Science) | PES University, Bengaluru — Aug 2020 to May 2024

Publications: “Synthesizing 3D Faces and Bodies from Text” (IEEE I2CT 2024); “Optimal IPL Playing 11 Team Selection” (IEEE I2CT 2023)